Classification: Introduction, Bayes Classifier and K-Nearest Neighbours

Course of Statistical Learning

Gareth et al.. An Introduction to Statistical Learning, Ch. 4.1 and 2.2.3

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Presentation Outline

Qualitative Variable

- In many situations, the **response variable** is **qualitative**: for example, eye color is qualitative, taking on blue, brown, or green, as possible outcome. Often qualitative variables are referred to as **categorical**.
- The process for **predicting qualitative responses** is known as **classification**. Predicting a qualitative response for an observation can be referred to as classifying that observation, since it involves **assigning the observation to a category, or class**.
- Many of the **concepts** encountered in **regression**, such as the bias-variance trade-off, **transfer** over to the **classification** setting with only some modifications due to the fact that y_i is no longer numerical.

Classification

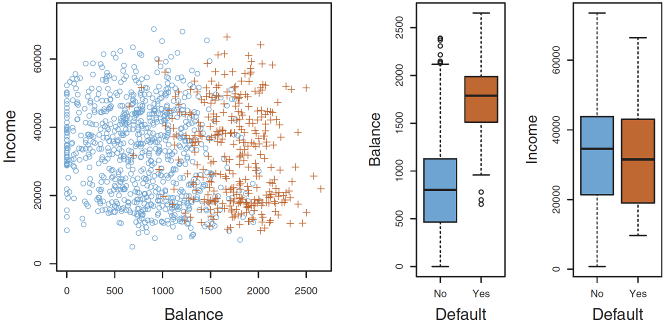
- Often the **methods used for classification** first **predict the probability** of each of the categories of a qualitative variable, as the basis for making the classification.
- There are **many possible classification techniques**, or classifiers, that one might use to predict a qualitative response. Three of the **most widely-used** classifiers are:
  - **K-nearest neighbours**
  - **logistic regression**
  - **Linear Discriminant Analysis (LDA)**
- Just as in the regression setting, we have a set of **training observations** $(x_1, y_1), \dots, (x_n, y_n)$ that we can use to build a classifier, that has to perform well not only on the training data, but also on an independent **test set**.

Classification – Examples

Some examples include:

1. A person arrives at the emergency room with a set of **symptoms** that could possibly be attributed to one of three **medical conditions**. Which of the three conditions does the individual have?
2. **Fraud detection**: an online banking service must be able to determine whether or not a transaction is **fraudulent**, on the basis of the user's **IP address, past transactions history and so forth**.
3. On the basis of DNA sequence data for a number of patients with and without a given **disease**, a biologist would like to figure out which **DNA mutations** are deleterious (disease-causing) and which are not.

Predicting whether an individual will default



Based on the **annual income** and the **monthly credit card balance**, the plot shows individuals who **defaulted** in a given month (in **orange**), and those who **did not** (in **blue**). Individuals who defaulted have **higher credit card balances** than those who did not. **Boxplots** show relations: balance - default status, and income - default status.

The Classification setting

- Suppose we seek to **estimate the classification function** f on the basis of training observations $\{(x_1, y_1), ..., (x_n, y_n)\}$ with $y_1, ..., y_n$ qualitative
- The **accuracy** of our estimate $\hat{f}$ can be quantified by the **training error rate**

$$\varepsilon_{train} = \frac{1}{n} \sum_{i=1}^{n} I(y_i \neq \hat{y}_i)$$

where $\hat{y}_i$ is the **predicted class label** and I is an **indicator function** (1 if $y_i \neq \hat{y}_i$, zero otherwise). ε_{train} is the fraction of incorrect classifications

- The **test error rate** is computed from predictions not used for training

The Bayes Classifier

- The **test error rate** is **minimized**, on average, by a very simple **classifier** that assigns each observation to the **most likely class, given its predictor values**.
- More precisely, we should simply **assign** a test observation with predictor vector x_0 **to the class** j for which

$$Pr(Y = j \mid X = x_0)$$

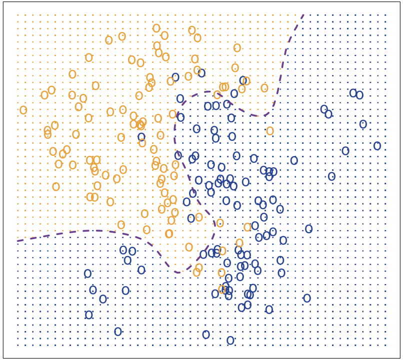
  is the **largest**. Pr denotes the **conditional probability**, i.e. the probability that $Y = j$, given the observed predictor vector x_0.
- This very simple classifier is called the **Bayes classifier**.
- In a **two-class problem** (only two possible response values, *class 1* or *class 2*), the Bayes classifier corresponds to predicting class 1 if $Pr(Y = 1 \mid X = x_0) > 0.5$, and class 2 otherwise.

The Bayes decision boundary

Simulated dataset. **Orange** and **blue** circles: training observations of **two different classes**. For each value of X_1 and X_2, there is a different **probability of the response being orange or blue**

Orange region: points for which $Pr(Y = orange \mid X) > 0.5$. **Blue region**: points for which $Pr(Y = orange \mid X) < 0.5$. Dashed line: **Bayes decision boundary** representing points where the probability is exactly 0.5

X_2



X_1

The Bayes decision boundary

- The **Bayes classifier's prediction** is determined by the **Bayes decision boundary** (i.e., observations on the orange side of the boundary are assigned to the orange class, observations on the blue side assigned to the blue class)

- **The Bayes classifier produces the lowest possible test error rate**, called the **Bayes error rate**.

- The Bayes classifier chooses the class with the largest conditional probability, hence the **error rate at** $X = x_0$ **is** $1 - \max_j Pr(Y = j \mid X = x_0)$

- In general, the **overall Bayes error rate** is given by

$$1 - \mathbb{E}\left(\max_j Pr(Y = j \mid X)\right)$$

where the expectation averages the probability **over all possible values of** X

Presentation Outline

K-Nearest Neighbours

- **Problem:** For **real data**, we do **not know the conditional distribution of** Y **given** X, and so **computing the Bayes classifier is impossible**
- Therefore, the Bayes classifier serves as an **unattainable standard** against which to compare other methods
- Many approaches attempt to **estimate the conditional distribution of** Y **given** X, and then classify a given observation to the class with **highest estimated probability**
- Among this approaches, one of the **most used** method is the **K-nearest neighbours (KNN) classifier**

K-Nearest Neighbours

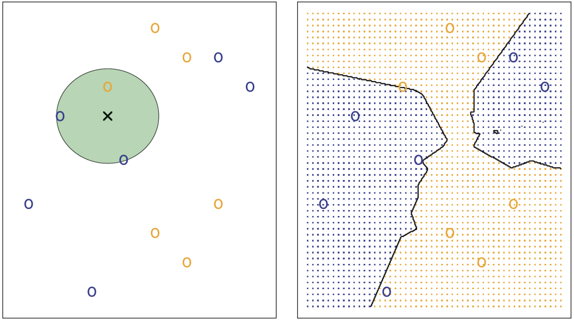
- Given a positive integer K and a test observation x_0, the KNN classifier first **identifies the K points in the training data that are closest to x_0** (symbol $\mathcal{N}_0$). This set of points represents the **neighbourhood**

- It then **estimates the conditional probability for class** j as the **fraction of points in $\mathcal{N}_0$ whose response values equal** j:

$$Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

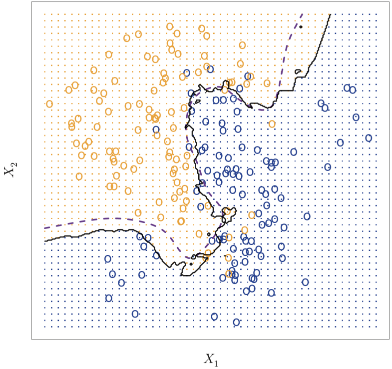
- Finally, *KNN* **classifies the test observation x_0 to the class with the largest probability**

Example of KNN approach: $K = 3$



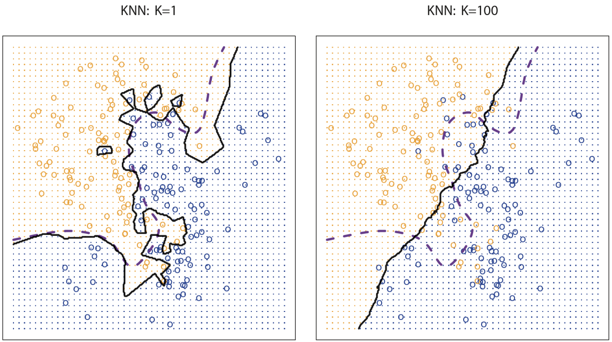
Goal: predict the class of the point labelled by the black cross. *KNN* will first **identify the three observations closest to the cross**: two blue points (estimated probabilities of 2/3 for the blue class) and one orange point, (1/3 for the orange class). Hence, KNN will **predict** that the **black cross belongs to the blue class**. The **KNN decision boundary** can be drawn by evaluating KNN to **all possible points** (X_1, X_2).

Largest dataset, $K = 10$



KNN is a very **simple approach** but it can produce classifiers **close to the optimal Bayes**. **Here**, the KNN decision boundary is very close to that of the Bayes classifier. The **test error rate** is 0.1363, which is close to the **Bayes error rate** of 0.1304.

Largest dataset, $K = 1$ vs $K = 100$

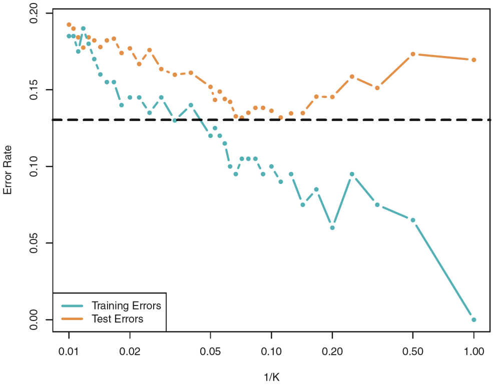


$K = 1$: the **decision boundary is overly flexible** (patterns not corresponding to the Bayes decision boundary). It corresponds to a classifier with **low bias** and **very high variance**. As K **grows**, the method becomes **less flexible** and produces a decision boundary that is close to linear, that corresponds to a **low-variance** but **high-bias** classifier.

Train and Test error in KNN

- As in the regression setting, there is not a strong relationship between the training error rate and the test error rate. With $K = 1$, the **KNN training error rate is 0**, but the **test error rate may be quite high**.
- If we use **less flexible classification** methods, the **training error rate** will **increase** but the **test error rate may decrease**.
- Choosing the **correct level of flexibility** is **critical**. The **bias-variance tradeoff**, and the resulting **U-shape** in the test error, can make this a difficult task.
- As $1/K$ increases, the method becomes **more flexible**. The training error rate consistently declines as the flexibility increases. However, the test error exhibits a characteristic U-shape, showing an increased error when the method becomes excessively flexible and overfits

KNN test and train errors as a function of $1/K$



The KNN training error rate (200 observations) and test error rate ($5,000$ observations) for a simulated dataset as a function of the level of flexibility (assessed using $1/K$). The black dashed line indicates the Bayes error rate. The jumps are due to the small size of the training data set.